K-mean clustering

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SETTING A WORKING DIRECTORY

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

load libraries

```
library(cluster)
library(factoextra)
library(dplyr)
library(readxl)
```

load data

```
data <- read_excel("mall_customers.xlsx")
head(data)
```

```
# A tibble: 6 × 6
  Customer_ID   Age Annual_Income Spending_Score Years_As_Member
        <dbl> <dbl>         <dbl>          <dbl>            <dbl>
1           1    19            15             39                1
2           2    21            15             81                2
3           3    20            16              6                1
4           4    23            16             77                3
5           5    31            17             40                2
6           6    22            17             76                1
# i 1 more variable: Average_Purchase_Value <dbl>
```

structure of data

```
str(data)
```

```
tibble [120 × 6] (S3: tbl_df/tbl/data.frame)
 $ Customer_ID           : num [1:120] 1 2 3 4 5 6 7 8 9 10 ...
 $ Age                   : num [1:120] 19 21 20 23 31 22 35 23 64 30
...
 $ Annual_Income         : num [1:120] 15 15 16 16 17 17 18 18 19 19
...
 $ Spending_Score        : num [1:120] 39 81 6 77 40 76 6 94 3 72 ...
 $ Years_As_Member       : num [1:120] 1 2 1 3 2 1 4 2 5 1 ...
 $ Average_Purchase_Value: num [1:120] 45.5 120 15.2 98.4 55 ...
```

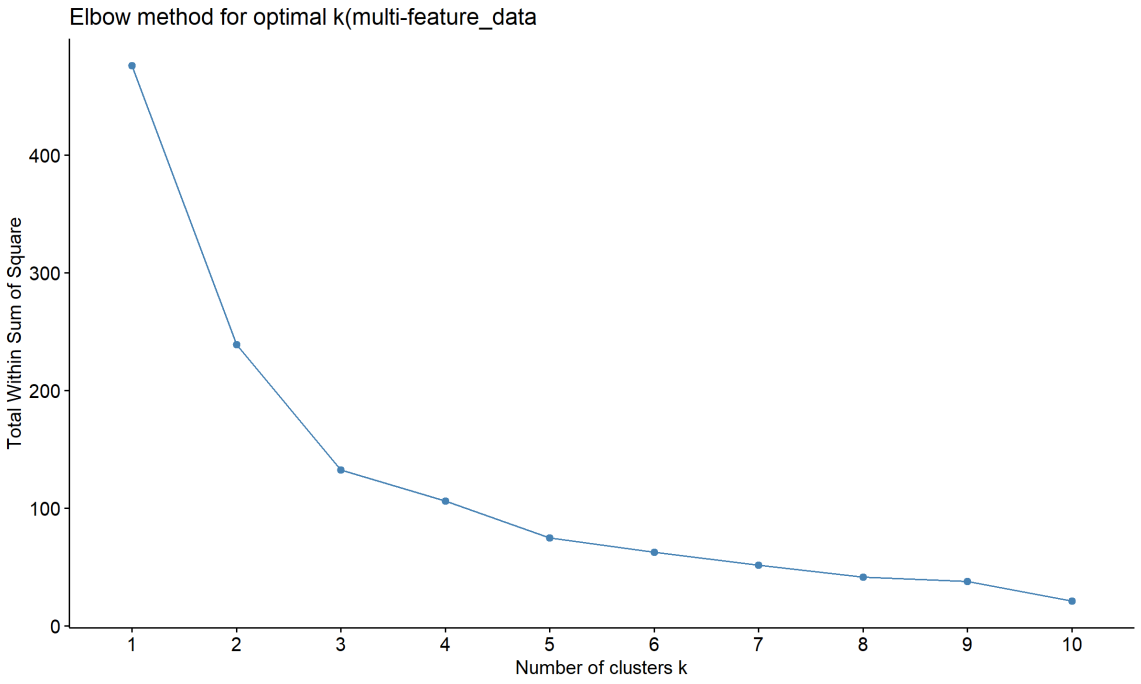
select numerical features for clastering

```
features <- data[, c("Age", "Annual_Income", "Spending_Score", "Averag
e_Purchase_Value")]
```

```
scaled_feature <- scale(features)
```

determine the optimal number of clusters

```
fviz_nbclust(scaled_feature, kmeans, method = "wss") + labs(title = "E
lbow method for optimal k(multi-feature_data")
```



apply k-means clustering(setting k=3 based on typical clusters pattern)

```
set.seed(42)
k_optimal <- 3
kmean_results <- kmeans(scaled_feature, centers = k_optimal, nstart =
25)
```

view cluster sizes and centers

```
print(kmean_results)
```

```
K-means clustering with 3 clusters of sizes 32, 56, 32

Cluster means:
         Age Annual_Income Spending_Score Average_Purchase_Value
1 -0.4886921    -0.8090175     -0.7155406             -0.7648435
2 -0.5409383    -0.1221158      0.9853105              0.9496514
3  1.4353342     1.0227202     -1.0087528             -0.8970464

Clustering vector:
  [1] 1 2 1 2 1 2 1 2 3 2 3 2 3 2 3 1 2 1 2 1 2 1 2 3 2 3 2 3 2 3 1 2
1 2 1 2 1
 [38] 2 3 2 3 2 3 2 3 1 2 1 2 1 2 1 2 3 2 3 2 3 2 3 1 2 1 2 1 2 1 2 3
2 3 2 3 2
 [75] 3 1 2 1 2 1 2 1 2 3 2 3 2 3 2 3 1 2 1 2 1 2 1 2 3 2 3 2 3 2 3 1
2 1 2 1 2
[112] 1 2 3 2 3 2 3 2 3

Within cluster sum of squares by cluster:
[1] 27.37949 73.82504 31.30489
 (between_SS / total_SS =  72.2 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.w
ithinss"
[6] "betweenss"    "size"         "iter"         "ifault"
```

append the clusters assignments back to the origina l data frame

```
data$clusters <- as.factor(kmean_results$cluster)
head(data)
```

```
# A tibble: 6 × 7
  Customer_ID   Age Annual_Income Spending_Score Years_As_Member
        <dbl> <dbl>         <dbl>          <dbl>            <dbl>
1           1    19            15             39                1
2           2    21            15             81                2
3           3    20            16              6                1
4           4    23            16             77                3
5           5    31            17             40                2
6           6    22            17             76                1
# i 2 more variables: Average_Purchase_Value <dbl>, clusters <fct>
```

visualize the clasters

```
fviz_cluster(kmean_results, data = scaled_feature, pallete = c("red",
"blue", "yelow"),
    geom = "point", ellipse.type = "convex", ggtheme = theme_minimal
(), main = "Clusters(5- Feature model visualized via PCA)")
```

